



USN

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Seventh Semester B.E. (CSE) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21CS701/20CSE81/16CSE81 - BIG DATA ANALYTICS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

	Unit – I	Marks	BT*	CO*	PO*
1. a) Describe the differences between semi-structured, structured, and unstructured data.		10	L*2	1	1
b) Explain the challenges of Big Data.		10	L2	1	2
2. a) Compare and contrast Real Time and Non Real Time Requirements.		10	L2	1	2
b) Explain with a diagram the cycle of Big Data Management.		10	L2	2	2
3. a) Explain the security Layer of Big Data Management.		10	L2	1	1
b) Explain Big Data Architecture with a neat diagram.		10	L2	2	1
	Unit – II				
4. a) Describe graph-based data stores and key value stores.		10	L2	3	1
b) Discuss about the Replica set of Mongo DB in details.		10	L2	3	2
5. a) Demonstrate CAP theorem in Big Data Solution and schema-less Model with example.		10	L2	3	1
b) Describe the important features of NOSQL. Explain Auto - sharding and replication.		10	L2	3	2
6. a) Explain how partitioning and combining is done in MapReduce.		10	L2	4	2
b) Explain the Characteristics of Mongo DB and Couch DB.		10	L2	3	2
	Unit – III				
7. a) Describe the data processing of hiveQL.		10	L2	4	2
b) Explain the following: i) Zookeeper ii) Sqoop		10	L2	5	1
8. a) Explain in detail about basics of YARN tool/concept.		10	L2	4	1
b) Compare and Contrast Business Intelligence and Big Data Analytics.		10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
